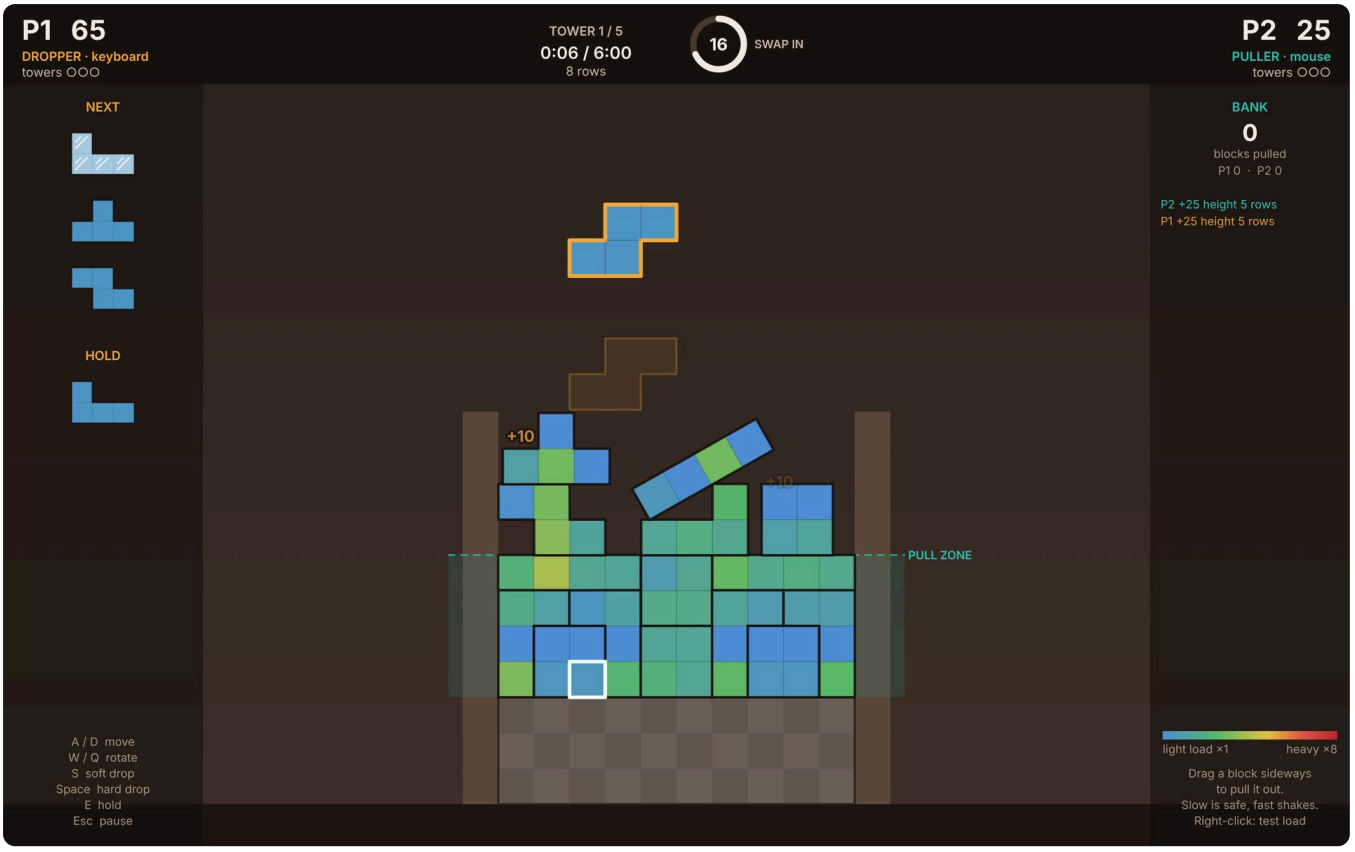
Counterweight

User guide for the prototype



Counterweight is a two-player block game that crosses Tetris with Jenga. Both players share one tower. One **drops** pieces onto the top. The other **pulls** blocks out of the bottom. Every 60 seconds you swap roles. When the tower falls, whoever caused it pays.

This guide covers the first prototype build: two players at one computer, sharing a keyboard and a mouse. It is an early test version, so expect rough edges.

1. Starting the game

You need a Windows PC or a Mac with a keyboard and a mouse, the game folder, and Python installed. The game runs in your web browser, but it has to be loaded through a small local server. Double-clicking `index.html` does not work.

Windows

1. Open the game folder in File Explorer and go into the `counterweight` folder.
2. Click the address bar at the top of the window, type `cmd` and press Enter. A black window opens.
3. Type this and press Enter:

   ```
   python -m http.server 8420
   ```

4. The window shows `Serving HTTP on :: port 8420` and then sits there. That is correct: it is running. Leave it open.
5. Open Chrome, Edge or Firefox and go to `http://localhost:8420/`

Mac

1. Open the Terminal app. Type `cd` and a space, drag the `counterweight` folder onto the window, and press Enter.
2. Type `python3 -m http.server 8420` and press Enter. Leave the window open.
3. Open your browser and go to `http://localhost:8420/`

> If you started the server one folder higher (in the folder that *contains* `counterweight`), the address is `http://localhost:8420/counterweight/` instead.

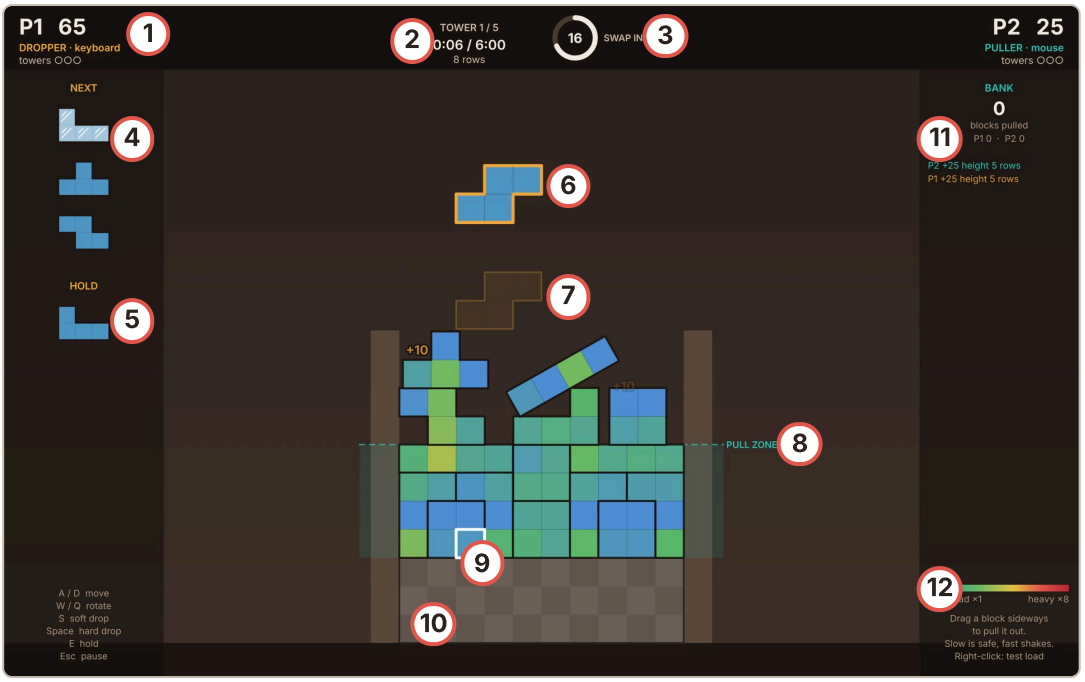
When the menu appears, press `Enter` or click **Start match**. To stop the server when you are done, close the black window.

2. The idea in one minute

- There are two players, P1 and P2, and two roles. P1 starts as the Dropper.
- The **Dropper** uses the **keyboard** to place falling pieces on top of the tower.
- The **Puller** uses the **mouse** to slide blocks out of the bottom half of the tower.
- Both play at the same time. Both score points from the same tower.
- Every 60 seconds the roles swap, and you trade the keyboard and the mouse.
- If the tower collapses, 150 points are taken away, split between the players according to who was to blame.
- The player with more points wins the tower. A match is the best of 5 towers.

The catch is that the tower you weaken as Puller is the one you have to build on a minute later, and the tower you build carelessly as Dropper is the one you will have to pull from.

3. The screen



#	What it is
1	**Player panel.** Score for this tower, current role and device, and towers won so far (filled dots). P2's panel is on the right.
2	**Tower status.** Which tower of the match this is, time played out of the 6-minute limit, and tower height in rows.
3	**Swap timer.** Seconds until the roles swap. It flashes for the last 10 seconds.
4	**Next.** The Dropper's next three pieces, top one first.
5	**Hold.** The piece the Dropper has set aside.
6	**Falling piece**, outlined in amber. The Dropper controls it.
7	**Landing shadow.** Where the piece would land if it dropped straight down now.
8	**Pull zone.** The shaded band below the dashed line. The Puller can only take blocks from here.
9	**Highlighted block.** The block under the Puller's mouse. A bright outline means it can be pulled.
10	**Foundation.** The grey base. It never moves and can't be pulled. The brown walls beside it hold in the first 8 rows; above them the tower stands free.
11	**Bank and log.** How many blocks have been pulled, and the most recent scoring events for both players.
12	**Load scale.** What the block colours mean (see section 6).

4. Playing the Dropper

You have the keyboard. Pieces appear above the tower and sink slowly. Steer each one, then let it land.

Key	Action
A D or ← →	Move left or right, one column at a time. Hold to keep moving.
W or ↑	Rotate clockwise.
Q	Rotate anticlockwise.
S or ↓	Soft drop: the piece sinks faster while you hold the key.
Space	Hard drop: let go of the piece now. It falls freely and lands harder the further it falls.
E	Hold: set this piece aside and take the held one (or the next one). Once per piece.
Esc	Pause or resume.

This is not quite Tetris

Once a piece touches the tower it becomes a real object with weight. It can tilt, slide, topple or knock other blocks. A piece that lands flat on a wide footing stays put. A piece balanced on one corner will tip over.

How the Dropper scores

- **+10** for each piece that is still resting on the tower 3 seconds after you place it.
- **+50** for completing a row. All 10 columns of a row must be filled, level and at rest. The row then **fuses** into one solid slab, drawn with dark bands. A slab is very stable and can't be pulled.
- **−10** if a piece you drop falls off the side without ever resting on the tower.

Special pieces

About one piece in eight is special.

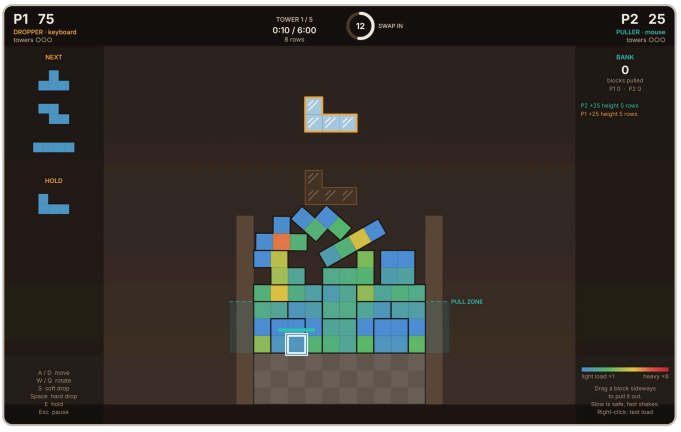
Piece	Looks like	Effect
Lead	Dark, with a square stud on each cell	Three times as heavy. Loads up whatever is underneath it.
Ice	Pale, with white streaks	Slippery. Things resting on or against it slide easily.
Cube	A single cell	A block the Puller took out, returned to you. Useful for patching gaps at the top.

The four loose rows the tower starts with never fuse. Only rows you build on top of them can.

5. Playing the Puller

You have the mouse. You take single blocks out of the pull zone, the shaded bottom half of the tower.

Mouse	Action
Point at a block	A bright white outline means you can pull it. A faint outline means you can't.
Hold the left button and drag sideways	Slides the block out. A bar above it shows progress. When the bar is full, the block is gone and you score.
Let go early	The block stays partly pulled. You, or the other player later, can carry on from there.
Right-click a block	Tap test: shows the block's exact load as a percentage. You then can't act for 0.75 seconds.



Pulling a block from the bottom row. The teal bar shows how far out it is; the label is from a tap test.

Speed matters

Drag slowly and the block comes out without disturbing anything. Yank it and the blocks touching it are dragged sideways too, more so if the block was heavily loaded. A full pull at a safe speed takes a little under two seconds.

What you can't pull

- Anything above the dashed pull-zone line. The line moves up as the tower grows, and is updated once a second.
- Fused slabs and the grey foundation.

How the Puller scores

- **10 points × the load multiplier** for each block: ×1 blue, ×2 green, ×4 yellow, ×8 red.
- **+5** for a clean pull, where none of the neighbouring pieces moved.

You pull one cell at a time, not a whole piece. The rest of the piece stays joined wherever its cells still touch edge to edge; if you take the middle out of a long piece, it becomes two separate pieces.

6. Load colours

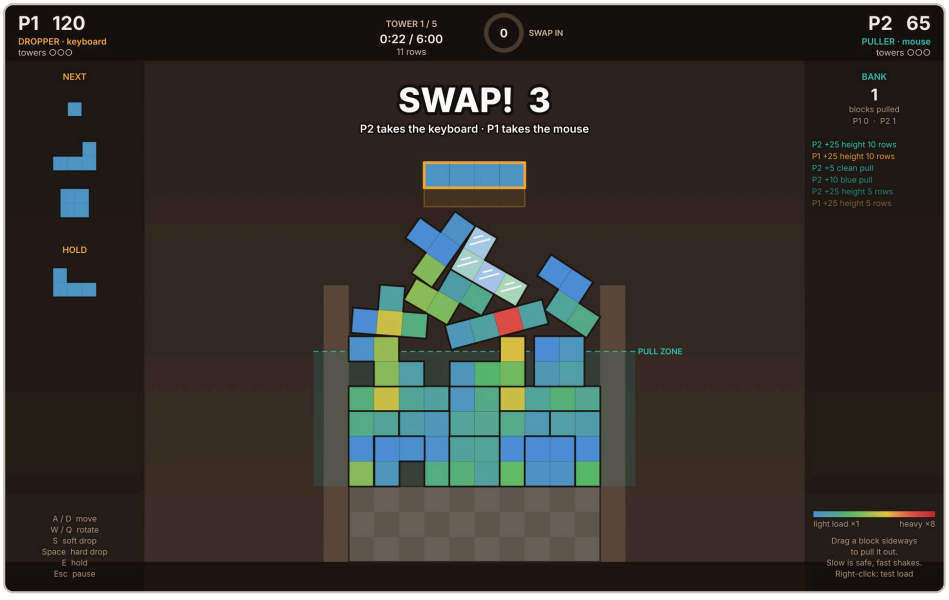
Every block is coloured by how much weight is passing through it, compared with its fair share of what sits above it. The colour tells the Puller two things at once: how risky the block is to remove and how much it is worth.

Colour	Load	Pull score	Meaning
Blue	under 30%	10	Carrying very little. Safe to take.
Green	30–60%	20	A normal share of the weight.
Yellow	60–80%	40	Carrying more than its share.
Red, pulsing	over 80%	80	Holding things up. Removing it may bring something down.

Colours shift as the tower changes. Taking a block moves its load onto its neighbours, so one pull can turn the blocks beside it yellow or red. Colours also flare briefly when a piece lands hard.

7. Swapping roles

- Roles swap every **60 seconds**. The timer ring flashes for the last 10.
- At the swap the game freezes for **3 seconds**. The tower keeps settling in slow motion, but nobody can act. Use this time to trade the keyboard and the mouse.
- The piece queue and the held piece carry over to the new Dropper.
- The colours follow the role: amber is always the Dropper, teal is always the Puller.



The swap freeze. The banner says who takes which device.

8. Collapse and blame

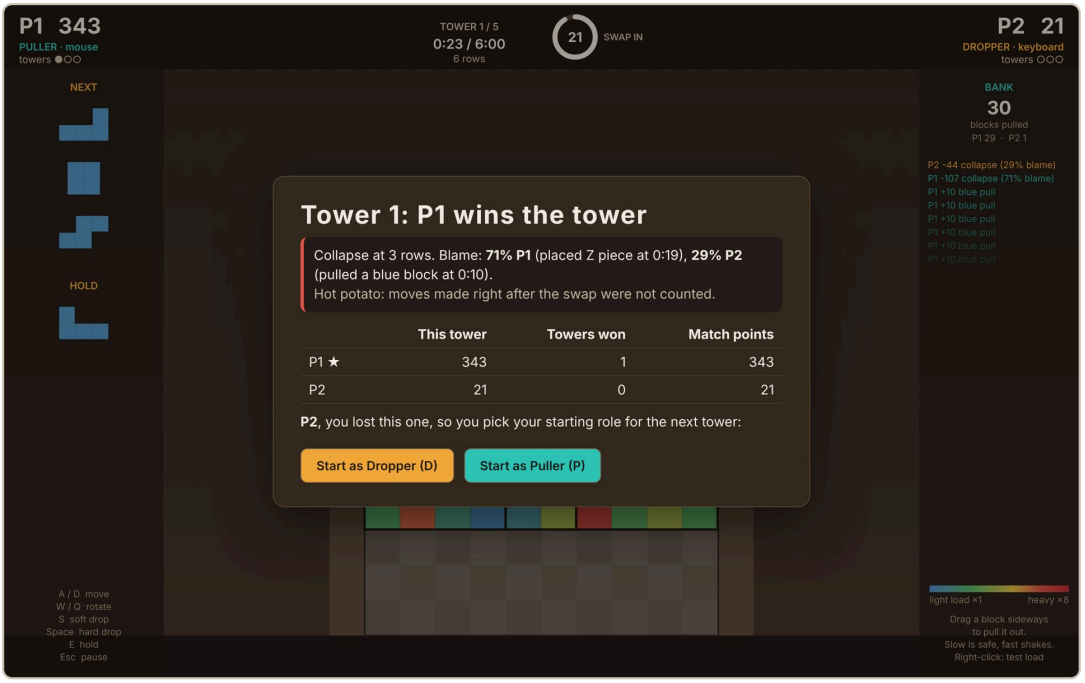
A **collapse** is when 10 or more blocks that had been resting fall more than two rows within a second and a half. Small sags, wobbles, and a single piece sliding off the side do not count.

When the tower collapses, the game looks back over the last 20 seconds and weighs up every move:

- Pulls count more than drops. Pulling a heavily loaded block or yanking it fast counts more still.
- Hard drops count more than gentle landings, and Lead pieces more than ordinary ones.
- Recent moves count more than old ones. A move's weight halves every 6 seconds.

The blame is shown as a percentage for each player, and the **150-point penalty** is split the same way.

> **Hot potato.** If the tower falls within a few seconds of a swap, moves made after the swap are ignored, so you can't be blamed for a tower that was already going when you took over. The window is 2 seconds for a tower up to 10 rows and grows by half a second for every 5 rows above that, up to 5 seconds.



The result screen after a collapse: the blame split with each player's most damaging move, the scores, and the role choice for the next tower.

If the tower is still standing after **6 minutes**, the tower ends there and both players get +100.

9. Scoring

Event	Who	Points
Piece placed and still resting after 3 seconds	Dropper	+10
Row completed and fused	Dropper	+50
Dropped piece falls off without ever resting	Dropper	−10
Block pulled: blue / green / yellow / red	Puller	+10 / +20 / +40 / +80
Clean pull (no neighbour moved)	Puller	+5
Tower reaches a new height of 5, 10, 15… rows	Both	+25 each
Tower survives the full 6 minutes	Both	+100 each
Collapse	Split by blame	−150 in total

Points always go to the player, not the role. Whatever you score as Dropper stays with you when you become the Puller.

10. Winning the match

- Whoever has more points when a tower ends wins that tower. If the points are level, the player with less blame wins it.
- The loser of a tower chooses their starting role for the next one: press `D` for Dropper or `P` for Puller, or click the button.
- A match is the **best of 5 towers**. It ends when one player has won 3, or after the fifth tower.
- If towers won are level at the end, the higher total of match points wins.
- Press `Enter` on the final screen to play again.

11. Tips

As Dropper

- Land pieces flat. Watch the landing shadow and avoid resting a piece on a single cell.
- Let pieces sink or soft-drop them onto uneven ground. Save hard drops for flat, solid spots.
- Fused rows are worth 50 and steady the tower, but they leave less for the Puller, and that will be you after the swap.
- Give the last piece time to settle before placing one on top of it.

As Puller

- Pull slowly. Speed gains you a second and risks the tower.
- Red blocks pay eight times as much as blue ones. Look at what is resting on one before you take it.
- Use the tap test when a block's colour is on the edge.
- Remember the blocks you remove come back to the Dropper as single cubes.

12. Troubleshooting

Problem	What to do
The black window seems stuck on `Serving HTTP on…`	That is normal. The server is running. Leave the window open and go to the browser.
`'python' is not recognized`	Python is not installed, or was installed without the **Add python.exe to PATH** box ticked. Run the installer from python.org again and tick that box.
The browser says the site can't be reached	Check the black window is still open and the address starts with `http://localhost:8420`
You see a list of files, or the Route Lens app	The server was started one folder too high. Add `/counterweight/` to the end of the address.
The page stays on "Loading physics..."	The page was probably opened by double-clicking the file. Open it through the `http://localhost` address instead.
An error about the port being in use	Use another number in both the command and the address, for example `8421`.
Keys do nothing	Click once on the game so the browser window has focus. Check the game is not paused (`Esc`).
Right-click opens a browser menu	Right-click inside the play area, not on the menus or outside the window.

Settings for test games

You can change a few settings by adding them to the end of the address. For example, `http://localhost:8420/?phase=20&cap=120` gives 20-second role phases and a 2-minute tower limit.

Setting	Meaning	Normal value
`phase`	Seconds between role swaps	60
`cap`	Tower time limit in seconds	360
`seed`	Any number. The same number gives the same order of pieces each time.	random

13. What this prototype does not have yet

- A computer opponent. You need two people, or one person working the keyboard and the mouse together.
- Online play.
- Sound.
- The slow-motion replay that shows how a collapse happened.
- Deep pulls from above the pull zone, and fused slabs cracking under load.
- The other special pieces: Glue, Spring, Ghost and Brace.
- The puzzle campaign.

The scoring values and timings in this guide are starting points and are likely to change after playtesting.